

Conor Glackin

Data Scientist | Machine Learning | Computational Biology

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Data scientist with 6+ years' experience developing statistical and machine-learning solutions across biological research and regulated pharmaceutical manufacturing. PhD in statistics-driven environmental microbiology with strong foundations in mathematics, computational biology and programming. Experienced building reproducible workflows, validating models, developing analytical tools and communicating complex findings to scientific, technical and non-technical stakeholders.

Key Achievements

- Developed an end-to-end machine-learning workflow for a **public-health early-warning system**, including dataset creation and processing, model development, validation and testing using Random Forest and LSTM.
- First author of “**AI-Driven Forecasting of *Vibrio vulnificus* in the Southern Baltic Sea Using High-Resolution Data**,” published in Water Research; co-authored a machine-learning benchmark across 13 biological datasets.
- Led analytics, automation and data-infrastructure projects in a regulated pharmaceutical manufacturing environment, including an internal analytics interface, SQL data-warehouse upgrade and automated reporting workflows.

Technical Skills

- **Programming:** R, Python, SQL, JMP/JSL, Bash, Git/GitHub, Linux
- **Machine learning and statistics:** Random Forest, LSTM/TensorFlow/Keras, regression, time-series analysis, design of experiments, cross-validation, model validation and performance evaluation
- **Scientific data:** Microbiome, environmental, healthcare/manufacturing, and genomic datasets
- **Data and delivery:** End-to-end machine learning workflows, reproducible pipelines, data visualisation, R Shiny, Power BI, ggplot2 and technical reporting

Work Experience

PhD researcher, Leibniz Institute for Baltic Sea Research, Rostock, Germany **Feb. 2022 – Aug. 2026**

- Developed and validated an early-warning system for pathogenic bacterial presence using Random Forest, time-lagged models and LSTM across ~1,500 samples and more than 2,000 variables.
- Built reproducible R and Python/TensorFlow pipelines for multi-source data integration, feature selection, model training, evaluation and visualisation.
- Designed spatial and temporal splits to test model generalisation across unseen locations, evaluating performance under severe class imbalance using ROC-AUC and AUPRC.
- Translated time-series modelling results into a practical public-health forecasting framework and presented findings at multiple international conferences.

Data Scientist, Abbott Laboratories, Donegal, Ireland **Nov. 2019 - Jan. 2022**

- Delivered statistical analysis, validation plans, development to production testing, sampling plans, root-cause investigations, SQL-supported reporting workflows, Power BI dashboards, and process optimisation projects in a regulated pharmaceutical manufacturing environment
- Conducted **comparative statistical analysis of clinical (patients' blood) and synthetic data** for glucose conversion factors, which was used for blood glucose testing during manufacturing
- Led development of the Smart New Analytics Project, an **internal user-facing analytics interface** used across multiple teams for reviewing data and reports, building automated reproducible data pipelines and data interfaces
- Led €30,000 **SQL data warehouse upgrade**, collaborating with the internal team and external vendors to implement improved data infrastructure
- **Automated** reporting workflows producing graphs and reports from product testing data (Excel files), reducing manual processing time from **8 hours to 2 minutes**
- Led **Lean Six Sigma Process Improvement project on predictive modelling** for on-site blood glucose strip testing using upstream variables as predictors

Mathematics and Statistics Teaching, University of Galway, Ireland **Sep. 2018 – Jul. 2019**

- Taught undergraduate students in mathematics, applied mathematics, statistics, and probability.

Education

PhD, **Statistics in Environmental Microbiology**,

Leibniz Institute for Baltic Sea Research, Rostock, Germany.

Feb. 2022 – Aug. 2026

Thesis: *Quantification, Prediction and Forecasting of Vibrio vulnificus in the Southern Baltic Sea Using High-Resolution Data*

MSc, **Computational Genomics**, University of Galway, Ireland.

Sep. 2018 – Aug. 2019

Thesis: *Predicting Links in Gene Regulatory Networks from other Networks*

Developed and compared statistical and machine learning models in R using repeated 10×10-fold cross-validation.

BSc, **Mathematics**, University of Galway, Ireland.

Sep. 2014 – May. 2018

Thesis: *Determinants of Wartime Success*

Selected Publications

- **Glackin, C.C.** et al. (2026). *AI-Driven Forecasting of Vibrio vulnificus in the Southern Baltic Sea Using High-Resolution Data*. Water Research
- Zschaubitz, E., **Glackin, C.C.** et al. (2025). *A Benchmark Analysis of Feature Selection and Machine Learning Methods for Environmental Metabarcoding Datasets*. Computational and Structural Biotechnology Journal
- **Glackin, C.C.**, et al. (2024). *Combined TCBS and CHROMagar Analyses Allow for Basic Identification of Vibrio vulnificus within a 48 h Incubation Period in the Coastal Baltic Sea*. Microorganisms

Further publications [Google Scholar Profile](#)

Additional Certifications & Skills

- Lean Six Sigma Yellow Belt [Abbott Diabetes Care, 2021]
- Advanced statistical software training (JMP)